CREDIT RISK
CLASSIFICATION

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INTRODUCTION

- We focused on building a model that predicts whether a borrower is likely to default on a loan using historical financial data, and demographical data.
- We used different models to identify borrower behavior patterns, and compare the models to which works the best.
- Default = 1; risky borrower (won't repay as agreed).
- Not default = 0; safe borrower

Real World Implication

- Helps banks reduce losses from unpaid loans
- Support smarter and faster loan approvals.
- Allow lenders to identify high-risk borrowers earlier.
- Impacts real people applying for mortgages, car loans, and most importantly student loans.
- Helps balance two goals: protecting lenders from losses, while still giving better access to loans.

MOTIVATION

- Poor loan decisions, can lead to significant loss and debt, for both banks and normal people.
- For normal people, it makes loan decision more clear, consistent, and accessible according to their budget.
- Helps avoid giving loans to people who may struggle to repay, and end up in debt.

DATASET OVERVIEW

Source: “Default of Credit Card Clients Dataset”
Link:https://www.kaggle.com/datasets/uciml/default-of-credit-card-clients-dataset

Content:

- 30,000 clients (rows)
- 23 features (columns)

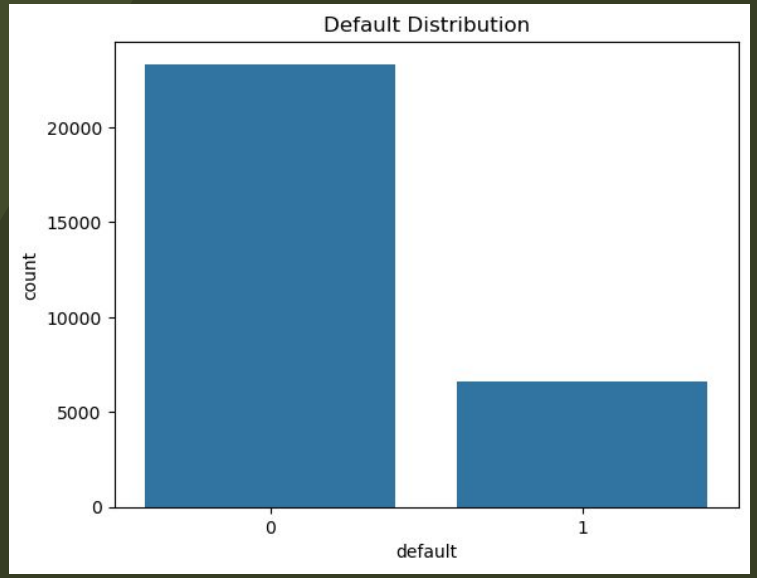
Target:

- 0 = No default
- 1 = Default

Examples of variables:

- Credit limit
- Payment history
- Bill amounts
- Demographics

EXPLORATORY DATA ANALYSIS

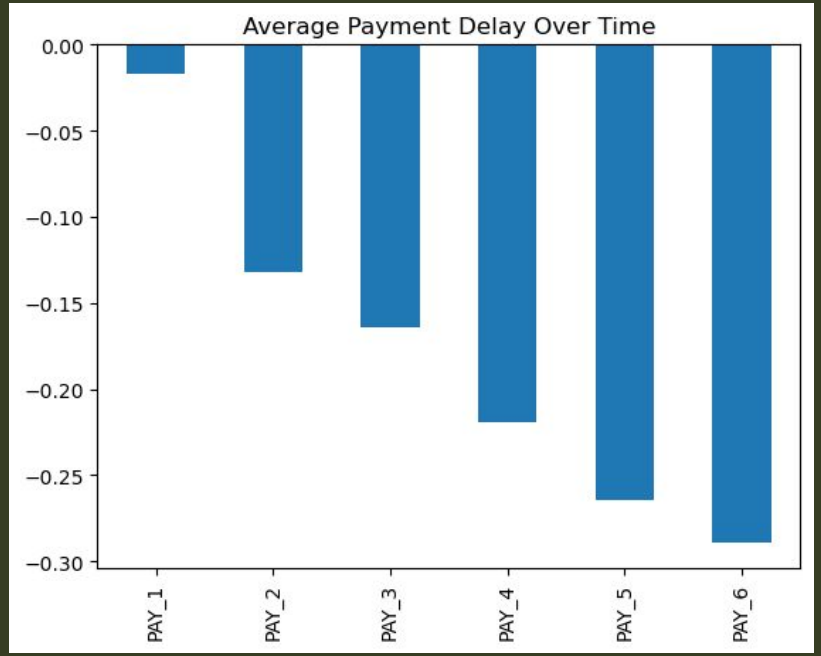


- The dataset is moderately imbalanced, with most borrowers (78%) classified as non-default (0) and approximately 22% classified as default (1).

- This class imbalance motivated the use of precision-recall analysis and threshold tuning during model evaluation.

EXPLORATORY DATA ANALYSIS

- Repayment status worsens from earlier to recent months, suggesting borrowers became less current on payments as time progressed.
- This trend indicates rising debt burden and increasing financial stress over time.



DATA CLEANING

- Verified **no missing values** across all features
- Merged ambiguous categories into a single **“other/unknown” group**
- No outliers were removed (Outliers may represent **high-risk customers**, which are important for prediction)

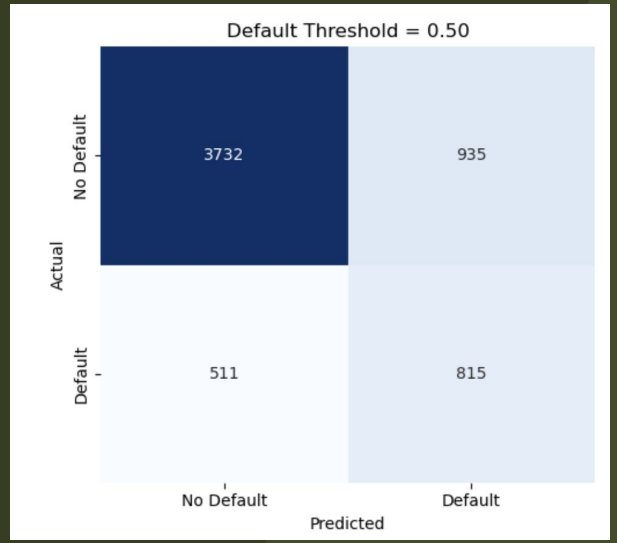
- Created **total bill amount** across months
- Created **total payment amount** across months
- Created ratios like:
  - **Pay ratio: total pay / total bill (signal)**
  - **Credit utilization: total bill / limit balance (debt to limit percentage)**

XGBOOST

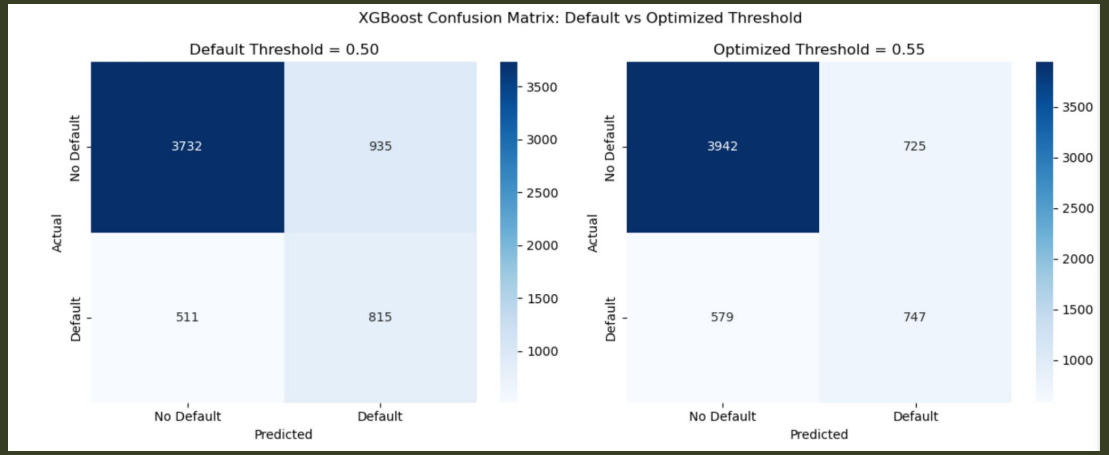
XGBOOST
Main Results

- Accuracy ≈ 76%
- Precision ≈ 46.6%
- Recall ≈ 61.5%
- F1 Score ≈ 0.53
- ROC-AUC ≈ 0.78
- Model balances detection of defaulters and false alarms
- Better performance than baseline models
- Confusion Matrix shows:
  - Good identification of non-defaulters
  - Still misses some risky borrowers

```
XGBoost Results - Default Threshold 0.50
========================================
Precision : 0.4657
Recall    : 0.6146
F1-Score  : 0.5299
ROC-AUC   : 0.7756

Confusion Matrix:
[[3732  935]
 [ 511  815]]

Classification Report:
              precision    recall  f1-score   support

           0       0.88      0.80      0.84      4667
           1       0.47      0.61      0.53      1326

    accuracy                           0.76      5993
   macro avg       0.67      0.71      0.68      5993
weighted avg       0.79      0.76      0.77      5993
```



XGBOOST RESULTS
Threshold Optimization



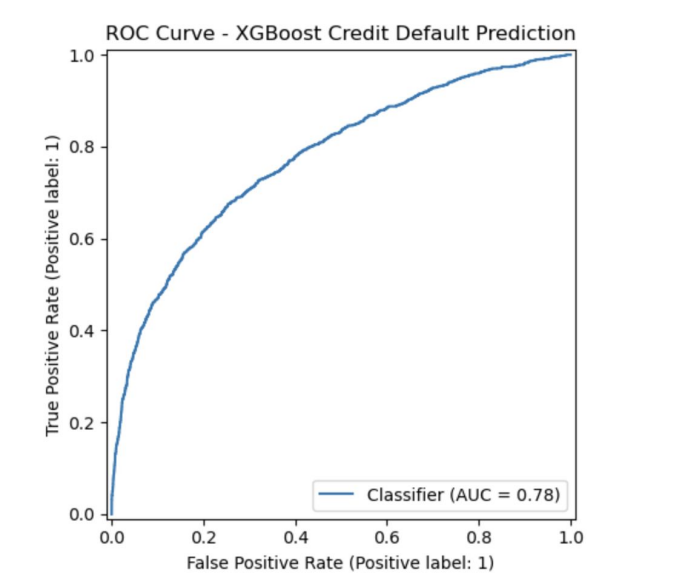
- Default threshold = 0.50
- Optimized threshold = 0.55

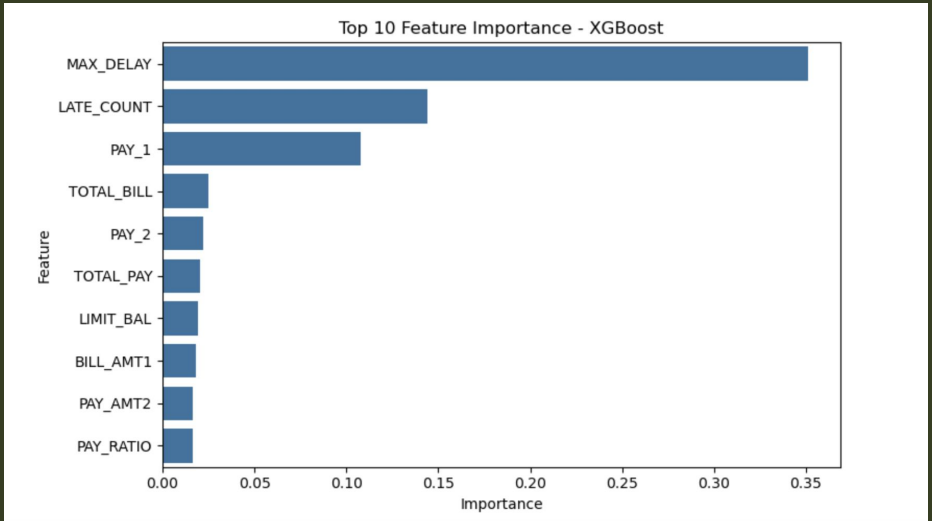
Changes after optimization:

- Precision ↑ (0.46 → 0.51)
- Recall ↓ slightly (0.61 → 0.56)
- F1 Score ↑ (0.53 → 0.54)
- Model becomes more precise
- Fewer false positives (less risky customers flagged incorrectly)
- Tradeoff:
  - Slightly fewer defaulters caught
  - But predictions are more reliable

XGBOOST RESULTS

- ROC-AUC ≈ 0.78
- Strong ability to distinguish:
- Defaulters vs Non-defaulters
- Model performs much better than random guessing
- Curve close to top-left = good performance





Top features:

- MAX_DELAY (most important), LATE_COUNT, PAY_1, TOTAL_BILL, PAY_2

Insights:

- Payment delays are the strongest predictors
- Financial behavior matters more than demographics
- Customers with frequent delays → high default risk

Naive Bayes

Naive Bayes

Main Results:

- Accuracy = only 35% correct
- Precision = only right 24% of the time.
- Recall = catches 90.42% of the actual defaulters
- F1 score is high but still terrible, combining recall and precision.
- Confusion matrix, wrongly calls a huge number of safe borrowers defaulters.
- CR: biased towards case 1

```
Naive Bayes Results
====================================
Accuracy : 0.3544
Precision: 0.2427
Recall   : 0.9042
F1 Score : 0.3826
ROC-AUC  : 0.7339

Confusion Matrix:
[[ 925 3742]
 [ 127 1199]]

Classification Report:
              precision    recall  f1-score   support

           0       0.88      0.20      0.32      4667
           1       0.24      0.90      0.38      1326

    accuracy                           0.35      5993
   macro avg       0.56      0.55      0.35      5993
weighted avg       0.74      0.35      0.34      5993
```

Naive Bayes

- Adjusting the threshold to 0.35, shows no real change.
- Cross Validated ROC-AUC
- Fold Score: 0.73 to 0.76
- Mean ROC-AUC: 0.7442
- Std-Dev: 0.0100
- The issue isn't that the model isn't learning. The issue is that its classification behavior is poor.
- We were expecting these results, as the financial variables in our dataset are dependent and not independent.
- Therefore, this is not a strong final model, but useful comparison baseline ML model.

```
Adjusted Threshold Results (threshold = 0.35)
========================================
Precision: 0.2393
Recall   : 0.9133
F1 Score : 0.3793

Adjusted Confusion Matrix:
[[ 818 3849]
 [ 115 1211]]

Cross-Validated ROC-AUC
========================================
Fold Scores: [0.75110294 0.73038796 0.73678916 0.7586016  0.74405222]
Mean ROC-AUC: 0.7442
Std Dev     : 0.0100
```

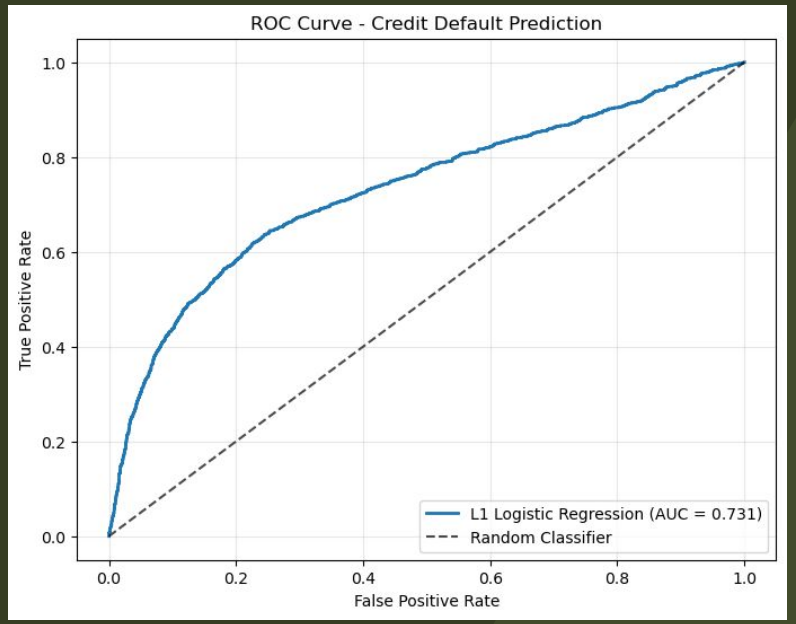
Logistic Regression

Logistic Regression

- Used Regularized Logistic Regression to classify borrower default risk.
- We chose this model due to its interpretable coefficients
- Strong benchmark for credit risk
- Probability outputs for threshold tuning

Model	ROC-AUC
L1 (Lasso)	**0.7313**
L2 (Ridge)	0.7312

- Final Selection:
  L1 Logistic Regression
  Because similar performance with simpler feature set.



Logistic Regression Performance Optimization

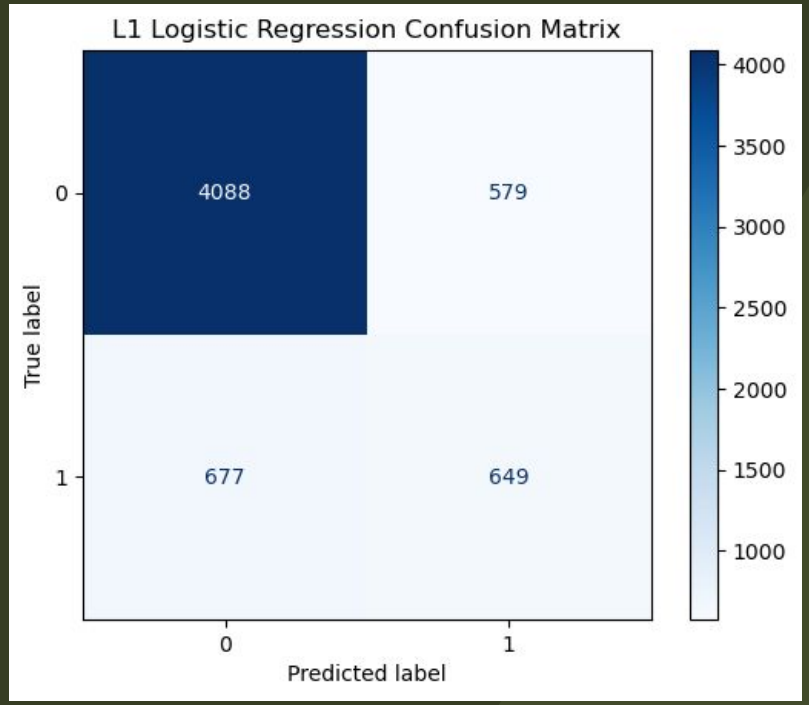
- Threshold Tuning

Adjusted classification threshold to improve default prediction quality.

- TN = 4087 (Borrowers who would not default, correctly classified as safe.)
- FP = 580 (Borrowers predicted to default, but actually would not.)
- FN = 677 (Borrowers predicted safe, but actually defaulted.)
- TP = 649 (Real defaulters correctly identified.)

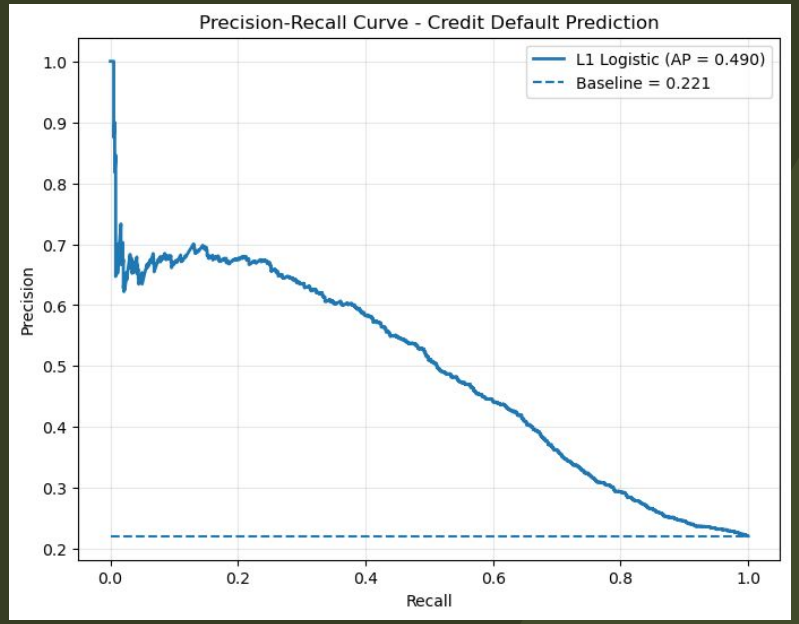
Threshold	Precision	Recall	Accuracy
0.50	44%	62%	0.74
0.60	**53%**	49%	0.79

- Improved precision by +9 percentage points
- Reduced false positive risk flags
- Better balance for lending decisions



Logistic Regression Precision–Recall Tradeoff

- Total test observations = 5993
- Defaults (class 1) = 649 + 677 = 1326
- So Baseline is: 1326 / 5993 = 0.221
- So random targeting would give ~22% precision.
- Used Average Precision (AP) to summarize Precision-Recall performance across all classification thresholds.
- Our model achieved 0.49, compared with a baseline of 0.221, indicating substantially stronger identification of risky borrowers.



Feed Forward
Neural Network

ANN Precision Recall Tradeoff

Feature Preprocessing

- Split data into training and testing sets using stratified sampling
- Applied StandardScaler to normalize features
- Converted data into PyTorch tensors for model training

Handling Class Imbalance

- Computed class imbalance ratio from training data
- Applied **weighted loss (pos_weight)** to emphasize minority class (defaults)

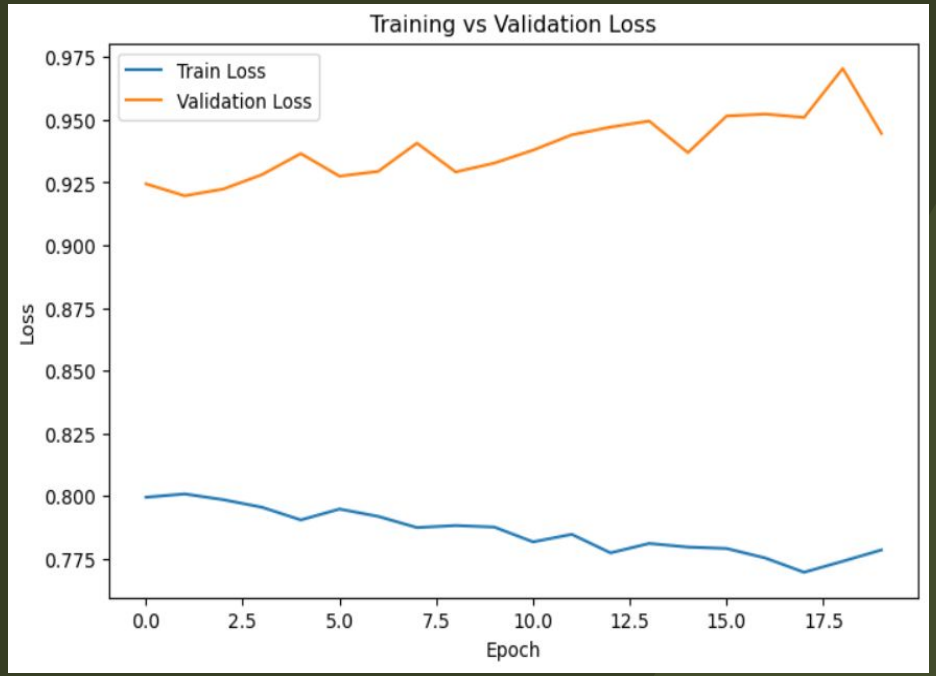
```
# handle class imbalance
num_pos = (y_train_tensor == 1).sum()
num_neg = (y_train_tensor == 0).sum()

pos_weight = num_neg / num_pos
```

✓ 0.0s

ANN Precision Recall Tradeoff

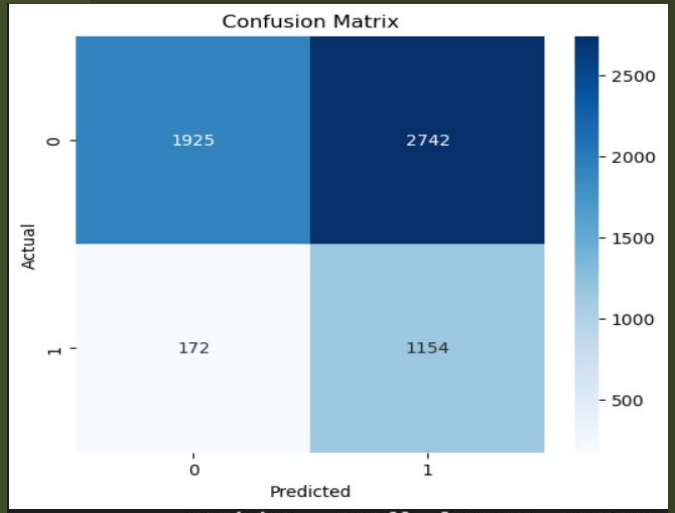
Model Architecture and Design

- Hidden layers: 2 layers (64 → 32 neurons)
- Activation: ReLU
- Batch size: 32 *(from training setup)*
- Epochs: 20 (baseline)
- Learning rate: 0.001
- Loss weighting: pos_weight for imbalance



ANN Precision–Recall Tradeoff

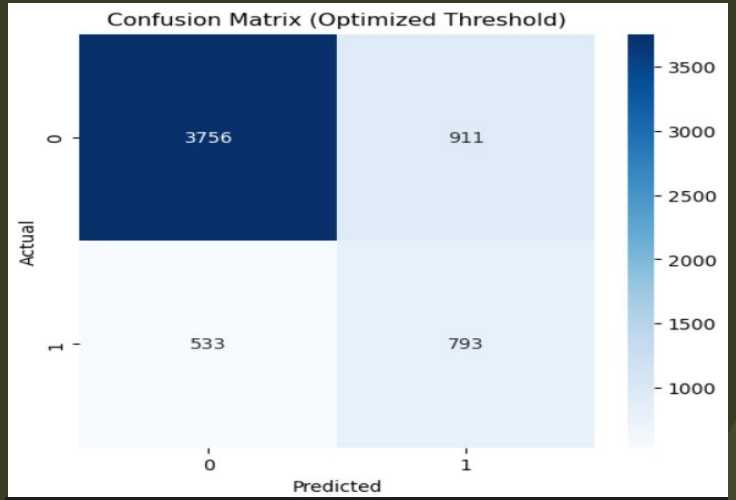
Threshold: 0.3



	precision	recall	f1-score	support
0.0	0.92	0.41	0.57	4667
1.0	0.30	0.87	0.44	1326
accuracy			0.51	5993
macro avg	0.61	0.64	0.51	5993
weighted avg	0.78	0.51	0.54	5993

Precision: 0.2962012320328542
Recall: 0.8702865761689291
ROC-AUC: 0.7664770874478584

F1 Optimized Threshold: 0.52



	precision	recall	f1-score	support
0.0	0.88	0.80	0.84	4667
1.0	0.47	0.60	0.52	1326
accuracy			0.76	5993
macro avg	0.67	0.70	0.68	5993
weighted avg	0.78	0.76	0.77	5993

Accuracy: 0.7590522275988654
Precision: 0.4653755868544601
Recall: 0.5980392156862745

CONCLUSION

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC	Key Insight
Naive Bayes	~35%	~24%	~90%	~0.38	~0.73	High recall but too many false positives (poor balance)
Logistic Regression	79%	53%	49%	~0.53	~0.73	Balanced and interpretable baseline
ANN (Neural Net)	~75–76%	~46%	~59%	~0.52	~0.76	Captures patterns but slightly overfits
XGBoost	**~78%**	**~51%**	**~56%**	**~0.54**	**~0.78**	**Best overall performance and balance**

In credit risk, missing a defaulter is more costly than a false alarm, so models were evaluated focusing on recall–precision tradeoffs.

THANK YOU